

ECON 0100 | Demo B | Solutions

Answers and work are shown in the red blocks.

Dragontear

Dragontear is a rare and powerful substance used in potions, spells, and fire-breathing, harvested from dragons while they laugh uncontrollably. Prices are in coins and quantity is in vials. The supply and demand equations for dragontear are known to be the following:

$$P = 60 - \frac{1}{10}Q_D \quad \text{and} \quad P = \frac{1}{5}Q_S$$

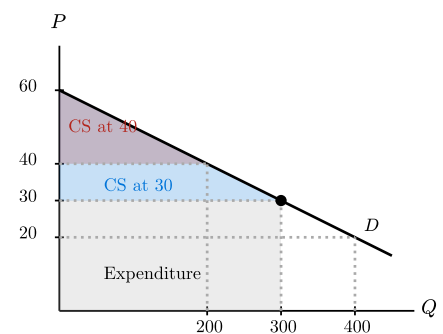
Q1 | Demand and Consumer Surplus (B1.1, B1.2)

Use a graph to plot this demand curve. Find and label the consumer surplus at a price of 30. On the same graph, show what happens to consumer surplus when the price increases to 40.

a) Quantity demanded at a price of 20 **400**, at 30 **300**, at 40 **200** (B1.1)

b) What is the marginal benefit of the 300th vial? **30** (B1.1)

c) CS at 30 **4500** CS at 40 **2000** Expenditure at 30 **9000** (B1.2)



Demand: $60 - Q/10$, vertical intercept 60, horizontal intercept 600.

a) $20 = 60 - Q/10$ gives $Q_d = 400$; at 30, **300**; at 40, **200**. Each step up in price moves along the curve to a smaller quantity.

b) $MB = 60 - 300/10 = 30$: the height of the demand curve at the 300th vial.

c) CS at 30 $= \frac{1}{2} \cdot 300 \cdot (60 - 30) = 4500$, the triangle below demand and above the price. At 40, $Q_d = 200$ and CS $= \frac{1}{2} \cdot 200 \cdot (60 - 40) = 2000$: the higher price cuts the triangle by 2500. Expenditure at 30 $= 30 \cdot 300 = 9000$, the rectangle under the price.

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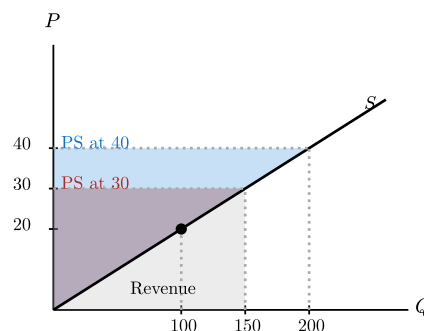
Q2 | Supply and Producer Surplus (B2.1, B2.2)

Use a graph to plot this supply curve. Find and label the producer surplus at a price of 30. On the same graph, show what happens to producer surplus when the price increases to 40.

a) Quantity supplied at 20 **100**, at 30 **150**, at 40 **200** (B2.1)

b) What is the marginal cost of the 100th vial? **20** (B2.1)

c) PS at 30 **2250** PS at 40 **4000** Revenue at 30 **4500** (B2.2)



Supply: $Q/5$, through the origin, so the first vial costs almost nothing to harvest.

a) $20 = Q/5$ gives $Q_s = 100$; at 30, **150**; at 40, **200**.

b) $MC = 100/5 = 20$: the height of the supply curve at the 100th vial.

c) PS at 30 = $\frac{1}{2} \cdot 150 \cdot 30 = 2250$, the triangle above supply and below the price. At 40, $Q_s = 200$ and $PS = \frac{1}{2} \cdot 200 \cdot 40 = 4000$: up by 1750. Revenue at 30 = $30 \cdot 150 = 4500$, the rectangle under the price; producer surplus is the part of it above the supply curve.

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Q3 | Price Elasticity (B5.2)

a) Use the midpoint method to find the elasticity of demand when the price changes from 10 to 20. $-1/3$

b) Use the midpoint method to find the elasticity of supply when the price changes from 20 to 40. **1**

c) Which of a)–b) is elastic, unit elastic, or inelastic? **a) inelastic, b) unit elastic**

a) Q goes from 500 to 400. Percent change in quantity over percent change in price, each measured from the midpoint:
 $(-100/450) \div (10/15) = (-2/9) \div (2/3) = -1/3$. Below one in absolute value: **inelastic**.

b) Q goes from 100 to 200. $(100/150) \div (20/30) = (2/3) \div (2/3) = 1$: **unit elastic**. A supply curve through the origin is unit elastic everywhere, since price and quantity always change in the same proportion.

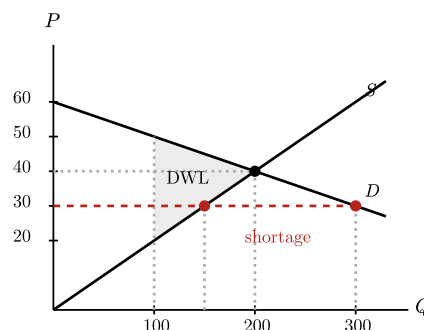
c) a) **inelastic**, b) **unit elastic**.

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Q4 | Equilibrium for Dragontear (B3.1, B4.1)

Use a graph to describe the market for Dragontear:

- a) Calculate equilibrium price: **40** Find equilibrium quantity: **200** (B3.1)
- b) At 30 coins: **shortage**, excess of size **150**, and the price will: **rise**, fall (B3.1)
- c) Calculate total surplus at equilibrium: **6000** (B4.1)
- d) What is the deadweight loss if only 100 vials are exchanged? **1500** (B4.1)



a) $60 - Q/10 = Q/5$ gives $60 = 3Q/10$, so $Q^* = 200$ and $P^* = 40$.

b) At 30, $Q_d = 300$ and $Q_s = 150$: a **shortage** of **150**. The price will **rise**: buyers who miss out bid it up, and sellers respond with more vials, until the gap closes at 40.

c) $TS = CS + PS = \frac{1}{2} \cdot 200 \cdot (60 - 40) + \frac{1}{2} \cdot 200 \cdot 40 = 2000 + 4000 = 6000$, or the whole triangle between the curves, $\frac{1}{2} \cdot 200 \cdot 60$.

d) At $Q = 100$, $MB = 50$ and $MC = 20$. The trades from 100 to 200 are worth making, since MB is above MC on each one, and their lost surplus is the triangle $DWL = \frac{1}{2} \cdot (50 - 20) \cdot (200 - 100) = 1500$.

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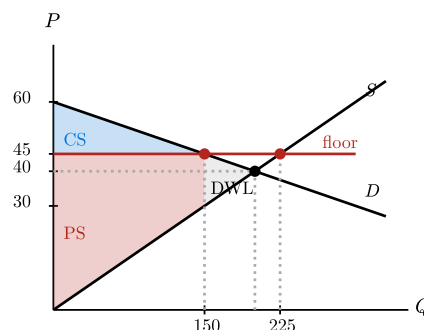
Q5 | Price Floor (B4.2)

Use a graph of the demand curve and the supply curve to evaluate the welfare effects of a price floor of 45.

a) CS **1125** Change in CS from Equilibrium **−875**

b) PS **4500** Change in PS from Equilibrium **+500**

c) DWL **375** Shortage/Excess **excess of 75**



The floor of 45 is above $P^* = 40$, so it binds. $Q_d: 45 = 60 - Q/10$ gives 150. $Q_s: 45 = Q/5$ gives 225. Only 150 vials are exchanged, the smaller of the two, leaving an **excess** of **75**.

a) $CS = \frac{1}{2} \cdot 150 \cdot (60 - 45) = \mathbf{1125}$, down **875** from 2000: buyers pay more and get fewer vials.

b) PS runs from the price down to supply over the 150 vials traded, where $MC = 30$: $\frac{1}{2} \cdot (45 + 15) \cdot 150 = \mathbf{4500}$, up **500** from 4000. The sellers who still trade gain more than the sellers who are shut out lose.

c) $DWL = \frac{1}{2} \cdot (45 - 30) \cdot (200 - 150) = \mathbf{375}$: the surplus on the 50 trades that no longer happen. Check: $6000 - 1125 - 4500 = 375$.

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Q6 | Second Wizarding War (B5.1, B5.3)

The second wizarding war limited the population of dragons. Use a graph to aid your discussion of the impact the war had on the market for dragontear.

a) The demand curve: shifted up, **stayed the same**, shifted down (B5.1)

b) The supply curve: **shifted up**, stayed the same, shifted down (B5.1)

In the years after the war, many more in the wizarding world required dragontear as part of a widespread rise in the popularity of spells. On its own (not including the change in the population of dragons), how has this change in popularity of spells changed the market for dragontear?

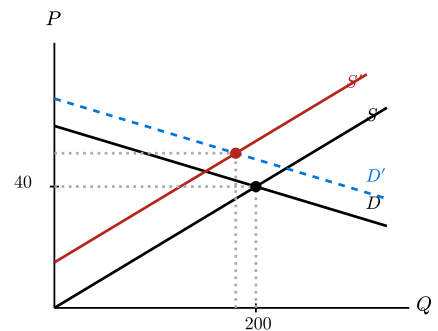
e) The demand curve: **shifted up**, stayed the same, shifted down (B5.1)

f) The supply curve: shifted up, **stayed the same**, shifted down (B5.1)

Without using numbers, how have these two changes together impacted the dragontear market equilibrium?

g) Equilibrium price has: **increased**, stayed the same, decreased, indeterminate (B5.3)

h) Equilibrium quantity has: increased, stayed the same, decreased, **indeterminate** (B5.3)



a) Demand **stayed the same**: the war changed the dragons, not what buyers are willing to pay.

b) Supply **shifted up**: with fewer dragons, each vial costs more to harvest, so sellers need a higher price at every quantity. On its own this raises the price and lowers the quantity.

e) Demand **shifted up**: more spells need dragontear, so buyers are willing to pay more at every quantity. f) Supply **stayed the same**.

g) Both shifts push the price the same way, so the price **increased**.

h) The shifts push quantity opposite ways: supply up lowers it, demand up raises it. Without the sizes of the shifts the change in quantity is **indeterminate**. The graph shows one case, where the supply shift is the larger of the two and quantity falls.